ConvNeXt AD/NC feasibility review

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Status: Retrospective review for a tutor consultation. The compulsory midpoint check-off has not been completed. This document does not establish prior approval; discuss an appropriate make-up arrangement with the tutor.

User need, scope and acceptance criteria

Investigate whether an imaging specialist can use a 2D MRI classification prototype to distinguish AD from NC and route uncertain predictions for review. Patient-level classification is the primary unit. The prototype is a research evaluation, with no established clinical deployment validity.

Patient separation	Zero shared patients/scans across train, validation and test; achieved by regrouping supplied data.
Accuracy requirement	ConvNeXt test accuracy at least 0.80. Final raw accuracy: slice 0.8818, scan 0.9270, patient 0.9118.
Baseline comparison	Same splits; compare class precision/recall, macro-F1, AUROC and resource cost. Patient macro-F1: final 0.9028 versus tuned baseline 0.7984.
Confidence routing	Project target: accepted accuracy at least 0.90 at coverage at least 0.50. Observed 0.9118 at coverage 1.0, but all nine errors accepted.
Resource constraint	Proposed project budget: peak allocated training VRAM below 4 GiB on one A100. Final observed peak 2,290.30 MiB; loop time 17.54 min.

Model choice and course concepts

Hard-difficulty ConvNeXt-Tiny: custom PyTorch modules, depths 3/3/9/3, channels 96/192/384/768, 27.82 million parameters. Transfer ImageNet-1K backbone weights to a one-channel input; compare with a 0.389-million-parameter CNN. Convolution, residual connections, normalization, cross-entropy, augmentation, regularization and validation-based selection connect to course concepts. References: Liu et al., A ConvNet for the 2020s (CVPR 2022); Guo et al., On Calibration of Modern Neural Networks (ICML 2017).

Data audit and measured feasibility evidence

ADNI provides 30,520 JPEG slices from 1,526 scans and 680 patients. Supplied folders overlap by 216 patients; pooled patient-stratified seed-3710 splitting gives 476/102/102 patients and 21,300/4,560/4,660 slices. Final test confusion matrix: 62 NC correct, 7 NC false positives, 2 AD false negatives, 31 AD correct. Baseline and both ConvNeXt recipes completed on one A100. Measured final batched forward latency is 0.4264 ms/slice, excluding loading and transfer.

Risks, budget, next experiment and fallback

Risks include correlated slices, only one seed, limited independent patients, acquisition shortcuts and confident errors. Two AD patients remain confidently missed by all three recipes. Test results were examined before fine-tuning, so a fresh cohort is needed for a confirmatory claim. The zero rejection cutoff demonstrates no useful uncertainty routing.

Proposed next budget: one A100 allocation up to 30 minutes to audit full-scan scores and intensity/position variation for existing errors; no further test-driven tuning. Test any quality-control hypothesis on validation data. Retain the cheaper CNN as a resource reference and require human review of prototype outputs. Multiple recipe changes mean the improvement cannot be attributed solely to architecture or pretraining.

Consultation record: Tutor/date: pending. Record actual feedback and any resulting changes after the discussion. Resource and confidence targets above are retrospective project choices, not assignment-specified thresholds.